

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

Program/s: B.Tech/MBA Tech

Year: II Semester: IV

Stream/s ; Data science

Subject: Introduction to data signal and image analysis

Time: 03 hrs (~~10:00am~~ to 1:00pm)

Date: 29 / 04 / 2023

No. of Pages: 03

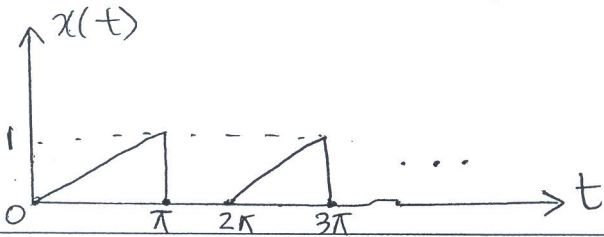
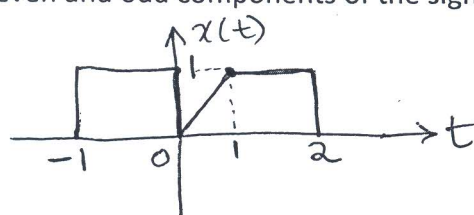
Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	marks									
CO-1 ; BL- 3	a.	If $X(S) = \frac{S^2}{(S^2+2S+2)(S+2)}$ then find $x(0)$ and $x(\infty)$	5									
CO-4 ; BL-1	b.	What is bit plane slicing? Find the bit planes of the given 3X3 image <table><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>0</td></tr><tr><td>7</td><td>2</td><td>1</td></tr></table>	1	2	3	4	5	0	7	2	1	5
1	2	3										
4	5	0										
7	2	1										
CO-4 ; BL-6	c.	5 x 5 Image is defined by the following expression $f(x,y)= x^2+y^2$ Construct the image, Determine the size of this image, Develop the image by retaining only row and column numbers 1, 3 and 5	5									
CO3- ; BL-2	d.	Obtain a discrete signal from $x(t) = 4 \cos 10\pi t$. Plot discrete time signal for unit and ramp function.	5									

Q2 CO1-; BL-3	a	Find FT for $x(t) = e^{at} \cos W_0 t$ for $t \geq 0$	10
Q2 CO2-; BL-5	b	Evaluate transfer function, impulse response and step response of a system given by the equation:- $y(n) - 3y(n-1) + 2y(n-2) = x(n)$	10
Q3 CO1-; BL-3	a	Determine trigonometric form of FS for the following waveform 	10
Q3 CO1-; BL-4	b	Write a detailed note on Time vs Frequency domain	10
Q4 CO-1; BL-3	a	If $X(S) = \frac{4(S+1)(S+3)}{(S+2)(S+4)}$ then find $x(t)$	10
Q4 CO1-; BL-2	b	Sketch the even and odd components of the signal 	10
Q5 CO32-; BL-2	a	Calculate $y(n)$ if $x(n) = \{1, 2, 3, 4\}$ and $h(n) = \{0, 2, 2, 2\}$ using Linear convolution. Enlist the different applications of convolution.	10
Q5 CO1-; BL-5	b	Perform Laplace for:- $L\left[\int_0^t e^{-2t} \cdot \frac{1}{t} \cdot \sin 2t \cdot \cos ht \cdot dt\right]$	10
Q6 CO1-; BL-3	a	Given $X(Z) = \frac{Z(Z+0.5)}{(Z-1)(Z-0.2)(Z-0.3)}$ find $x(n)$ for all possible ROCs.	10

Q6 CO2-; BL-1	b	What are FIR filters? Enlist the advantages of FIR Filters (compared to IIR filters)	10																				
Q7 CO4-; BL-4	a	What is histogram equalization? Perform histogram equalization on the following 4 x 5 image. The gray level distribution of the image is given below. Plot histogram of original image and equalized image. Comment on the results obtained after equalization. <table><tr><td>0</td><td>1</td><td>1</td><td>3</td><td>4</td></tr><tr><td>7</td><td>2</td><td>5</td><td>5</td><td>7</td></tr><tr><td>6</td><td>3</td><td>2</td><td>1</td><td>1</td></tr><tr><td>1</td><td>4</td><td>4</td><td>2</td><td>1</td></tr></table>	0	1	1	3	4	7	2	5	5	7	6	3	2	1	1	1	4	4	2	1	10
0	1	1	3	4																			
7	2	5	5	7																			
6	3	2	1	1																			
1	4	4	2	1																			
Q7 CO4-; BL-1	b	Define the following terms related to image fundamentals:- Image, Pixel, color image, color spaces, size of an image	10																				